

Knowledge Extraction from Autoencoder Latent Spaces: Diagnosing Effective Dimensionality via Active-Unit Dynamics and Intrinsic-Dimension Estimates

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Abstract

Determining how many latent coordinates an autoencoder (AE) actually uses is difficult because reconstruction error, intrinsic-dimension estimates, and variational activity reflect different aspects of a learned representation. This paper proposes a multi-indicator diagnostic framework for extracting actionable knowledge from AE and variational-autoencoder (VAE) latent spaces. The framework combines active-unit (AU) dynamics, TwoNN/MLE intrinsic-dimension (ID) estimates, reconstruction-error elbows, and PCA baselines, while explicitly separating their diagnostic roles: AU values as direct observations of coordinate usage; TwoNN/MLE as numerically stable auxiliary references subject to geometric bias; and the AU slowdown point, m^{knee} , as an exploratory, grid- and seed-dependent change point rather than a unique optimal bottleneck size. Under a linear-Gaussian β -VAE rate-distortion surrogate, we derive the activation condition $\lambda_j > \beta\tau^2$, providing an interpretable water-filling view of KL-induced dimension suppression. Experiments on fully connected VAEs trained on MNIST show that AU values ($\sigma \leq 0.5$) and TwoNN estimates (10.03 ± 0.12) are stable across three random seeds, whereas sparse-grid m^{knee} is unreliable (43 ± 15). A dense grid around the estimated intrinsic dimension stabilizes m^{knee} at 10.3 ± 0.9 , consistent with TwoNN near 10. Additional experiments on Conv-VAEs, CIFAR-10, and SVHN show that stable identification is not always possible and clarify the boundary conditions under which $\text{AU}/m^{\text{knee}}$ diagnostics should be regarded as inconclusive. These results support the proposed framework as a reproducible diagnostic protocol for latent-dimensionality analysis and model-design support, rather than a universal dimension estimator.

Keywords: variational autoencoder; effective latent dimensionality; active units; intrinsic dimension; TwoNN; diagnostic framework; representation learning; knowledge extraction; posterior collapse

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1. Introduction

Autoencoders (AEs) and variational autoencoders (VAEs) [11] learn compact representations of high-dimensional data through a bottleneck of dimension m . A central design question is: *how many of the m latent coordinates does the model actually use?* This “effective latent dimensionality” governs reconstruction fidelity, disentanglement, and posterior collapse [19].

From the viewpoint of machine learning and knowledge extraction, latent representations underpin model selection, compression, downstream prediction, clustering, and

visualization [7]. However, the nominal bottleneck dimension m does not necessarily coincide with the number of coordinates the model actually exploits. The objective of this study is not merely to estimate a single dimensionality value; rather, we aim to extract model-design knowledge from trained representations. The proposed diagnostics answer three practical questions: which latent coordinates are actually used by the model; whether the geometry of the learned representation is consistent with intrinsic-dimension estimates; and whether increasing the nominal bottleneck dimension leads to meaningful representational gains. This knowledge supports selecting latent-space sizes, detecting over-parameterized or under-regularized representations, and identifying cases where dimensionality diagnostics should be regarded as inconclusive. In this sense, the proposed framework provides a knowledge-extraction protocol for representation learning, rather than a standalone intrinsic-dimension estimator.

Existing work measures individual aspects of latent geometry—AU counts [18,19], ID estimates [8,9], and isometric regularization [12]—but a systematic framework that combines these indicators, clarifies their distinct roles, and defines the conditions under which each is reliable has not been established.

1.1. Contributions

The contributions of this paper are threefold.

1. **Multi-indicator diagnostic framework.** We propose a framework that explicitly assigns distinct roles to four indicator classes: (i) AU values as direct usage observations; (ii) TwoNN/MLE ID as numerically stable auxiliary references subject to geometric bias; (iii) m^{knee} as an exploratory change point; and (iv) MSE elbow and PCA as reconstruction-side baselines. The main contribution is the role separation itself, not m^{knee} as a new estimator. MNIST three-seed experiments confirm that AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12) are reproducible; the main stabilized m^{knee} result is 10.3 ± 0.9 under dense-grid, independent-initialization, multi-seed conditions (Section 5.4).
2. **Linear-Gaussian rate-distortion analysis.** Under a linear-Gaussian β -VAE rate-distortion approximation, we obtain the activation condition $\lambda_j > \beta\tau^2$ and show that AU saturation follows a water-filling principle. This is an interpretive approximation model, not a general theorem for non-linear VAEs (Analytical Result 1).
3. **Empirical characterization of reliability and limits.** Dense-grid/multi-seed experiments show that sparse-grid m^{knee} is unreliable ($\sigma = 15$), whereas dense-grid evaluation stabilizes it near the TwoNN estimate. Conv-VAE and natural-image experiments identify boundary conditions under which AU/ m^{knee} diagnostics become unreliable or require stronger regularization.

1.2. Paper Organization

Section 2 reviews related work. Section 3 defines the diagnostic framework. Section 4 provides theoretical background. Section 5 reports experiments. Section 6 discusses findings. Section 7 concludes.

2. Related Work

Representation learning and autoencoders.

Bengio et al. [7] survey representation learning, including the manifold hypothesis [6] underlying AE design. β -VAE [18] promotes disentanglement by raising the KL weight; Lucas et al. [19] analyze posterior collapse in the linear-VAE limit. Contractive AEs [20] and denoising AEs [21] provide complementary regularization strategies.

Intrinsic-dimension estimation.

TwoNN [8] and MLE [9] estimate local ID from nearest-neighbor distances; DANCo [14] offers higher accuracy at greater computational cost. Pope et al. [15] report $d_{\text{ID}} \approx 26$ for CIFAR-10 at 32×32 . Ansuini et al. [16] and Valeriani et al. [17] track ID across layers of deep networks and large transformers, respectively.

Geometric autoencoders.

Kato et al. [12] and Nazari et al. [13] propose isometric decoder-Jacobian regularization. We use IsometricAE as a controlled testbed for geometric bias, not as the main recommendation.

Positioning.

The present work differs from the above by targeting the *design stage*: combining AU dynamics and ID estimation to characterize effective dimensionality and quantify when each indicator is reliable or inconclusive.

3. Diagnostic Framework

3.1. Problem Setup

Let an AE/VAE have encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. For a VAE, the encoder outputs a posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$; KL regularization (β -weighted) acts as an information bottleneck. Reconstruction loss is mean-squared error (MSE) on $[0, 1]$ -normalized inputs.

Definition 1 (Active Units [19]).

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2}. \quad (1)$$

AU is stable for $\delta \in [10^{-3}, 10^{-1}]$ (Appendix E.1). Note: AU is defined for VAEs via the posterior mean. For standard AEs, we use a pseudo-AU based on encoder-output coordinate variance (all coordinates are typically active).

Definition 2 (AU Slowdown Point). Let $s(m_i) = (\text{AU}(m_i) - \text{AU}(m_{i-1})) / (m_i - m_{i-1})$ be the raw AU increase rate, and $\rho(m_i) = s(m_i) / \max_j s(m_j) \in [0, 1]$ the normalized rate. The AU slowdown point is

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\}, \quad \tau_{\text{knee}} = 0.25. \quad (2)$$

*m^{knee} is **not** a standalone dimension estimator; it is informative only under dense-grid, independent-initialization, multi-seed conditions.*

3.2. Indicator Role Separation

Table 1 summarizes the distinct roles assigned to each indicator. “Stable across seeds” means low inter-seed variance under the tested conditions, *not* unbiasedness toward the true ID. “Exploratory” means useful for hypothesis generation, not a unique estimate.

Table 1. Role classification and empirical reliability of diagnostic indicators.

Indicator	Category	Empirical Reliability	Primary Use
AU values	Stable across seeds[†]	$\sigma \leq 0.5$ (MNIST, 3-seed)	Direct observation of active coordinates
TwoNN/MLE ID	Numerically stable auxiliary[†]	$\sigma \leq 0.12$ ($m = 64$)	Reference ID estimate
MSE elbow	Auxiliary	Qualitative only	Reconstruction-side check
PCA cum. var.	Auxiliary	Linear assumption	Linear baseline
m^{knee}	Exploratory	$\sigma = 15$ (sparse); 0.9 (dense)	Rough heuristic; dense grid only

[†] Subject to isometry bias; numerically stable but not unbiased toward true d_{ID} . [‡] “Stable across seeds” means low inter-seed variance under tested conditions, not unbiasedness. m^{knee} requires dense-grid, independent per- (m, seed) initialization, and multiple seeds to be informative.

3.3. Two-Stage Diagnostic Pipeline

Stage 1 (auxiliary ID estimation). For synthetic data with known d_{true} , TwoNN/MLE is applied directly. For real data, a reference AE ($m_{\text{ref}} = 256$) is trained and its latent representation fed to TwoNN/MLE to obtain $\hat{d}_{\text{ID}}$. *Limitation:* the reference AE is not isometric ($\kappa \approx 10^9$ for MNIST); $\hat{d}_{\text{ID}}$ is an approximation subject to geometric bias.

Stage 2 (AU dynamics, primary). A VAE ($\beta = 4$) is trained on a grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$. For *sharp-saturation* data (synthetic), AU_{sat} directly indicates effective dimensionality. For *gradual-increase* data (real), m^{knee} (Equation (2)) is cross-checked against $\hat{d}_{\text{ID}}$ from Stage 1.

4. Theoretical Background

4.1. Linear-Gaussian Rate-Distortion Analysis

Theorem 1 (AU Activation Condition). *Consider a β -VAE with linear encoder/decoder and Gaussian prior, analyzed under a linear-Gaussian rate-distortion surrogate approximation (proof in Appendix B.2). Within this approximation, the water-filling argument gives:*

1. **(Activation condition)** The j -th eigen-direction becomes active iff

$$\lambda_j > \beta\tau^2, \quad (3)$$

where λ_j is the j -th data-covariance eigenvalue and τ^2 is the decoder noise variance.

2. **(Optimal active count)** $A^*(m) \leq \min(m, d^*(\beta))$, where $d^*(\beta) := |\{j : \lambda_j > \beta\tau^2\}|$.
3. **(Monotone saturation of optimal solution)** $A^*(m)$ is non-decreasing in m and saturates at $d^*(\beta)$.

Remark 1 (Scope). *Theorem 1(3) concerns the global-optimum count $A^*(m)$. In practice, non-linear VAEs trained with stochastic optimization yield an empirical $\text{AU}(m)$ that can be non-monotone due to finite-sample effects and initialization dependence (e.g., $\text{AU} = 29$ at $m = 64$ vs. 26 at $m = 128$, Table A1). This analytical result serves as an interpretive approximation model, not a theorem applicable to arbitrary VAE training.*

Remark 2 (Vanilla AE corresponds to $\beta = 0$). *Without KL regularization, the threshold $\beta\tau^2$ vanishes. The pseudo-AU of a standard AE thus approximates m .*

4.2. Grid-Resolution Dependence of m^{knee}

Proposition 1 (Grid-Resolution Dependence). *Under an idealized step-function AU model, $|m^{\text{knee}} - d^*(\beta)| \leq \max_i(m_{i+1} - m_i)$.*

Remark 3. This bound applies to the idealized model only. The practical implication is qualitative: a finer grid near $\hat{d}_{\text{ID}}$ reduces m^{knee} variance (Section 5.4).

4.3. Pullback Metric and Isometry

Proposition 2 (Local Isometry Condition). Decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ is locally isometric at z iff $G(z) = J_g(z)^\top J_g(z) = I_m$ (all singular values equal 1). The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ measures anisotropy: $\kappa = 1$ means all singular values are equal, but not necessarily 1. Therefore, κ is an anisotropy indicator, not a sufficient condition for isometry.

4.4. Design Principle: Bottleneck Margin is a Major TwoNN Stabilizing Factor

Design Principle 1. Under the tested conditions, the numerical stability of TwoNN estimates is explained more consistently by bottleneck margin ($m \gg \hat{d}_{\text{ID}}$) than by isometric regularization alone. TwoNN underestimates when $m \approx d_{\text{true}}$; it converges once $m \geq 5\hat{d}_{\text{ID}}$ approximately.

This is an empirically supported guideline verified primarily on MNIST multi-seed experiments and qualitatively checked on Fashion-MNIST. It should not be interpreted as a general claim that isometric regularization is irrelevant.

5. Experiments

5.1. Experimental Setup

Common settings. Adam ($\text{lr} = 10^{-3}$), batch 128, ReLU activations, Sigmoid output, MSE reconstruction loss on $[0, 1]$ -normalized inputs, PyTorch 2.0, seeds $\{42, 123, 777\}$. Each (m, seed) pair is initialized independently.

Synthetic manifolds. FC network $[N \rightarrow 64 \rightarrow 32 \rightarrow m]$, 200–300 epochs, VAE $\beta = 4$, $n = 5000$.

MNIST. $N = 784$, FC $[784 \rightarrow 512 \rightarrow 256 \rightarrow m]$, 300 epochs, $n_{\text{train}} = 20,000$. Grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$.

Fashion-MNIST (preliminary check). Same architecture, 100 epochs, $n_{\text{train}} = 10,000$, single seed only; used as a preliminary external check, not as evidence of seed-level robustness.

ID estimation: TwoNN ($k = 2$), MLE ($k = 10$), AU threshold $\delta = 10^{-2}$.

5.2. Synthetic Manifolds: Principle Verification

5.2.1. Reconstruction-Error Elbows

Table 2 shows AE reconstruction MSE on Swiss Roll ($d_{\text{true}} = 2$) and Torus ($d_{\text{true}} = 2$, $d_{\text{emb}} = 3$). For Swiss Roll, the MSE decreases substantially from $m = 1$ to $m = 2$, consistent with a two-dimensional unfolded parametrization. For Torus, the largest decrease occurs at $m = 3$, consistent with the three-dimensional embedding requirement (the Torus cannot be represented globally in $\mathbb{R}^2$ without topological distortion; a smooth standard embedding requires $\mathbb{R}^3$). The anomalous increase at $m = 4$ for Swiss Roll (0.041 ± 0.015 , higher variance than at $m = 2$) illustrates that MSE elbows are affected by optimization variance, decoder parametrization, and network capacity, not only by intrinsic dimensionality. Note that simple connectedness alone does not guarantee that the embedding dimension equals the intrinsic dimension. Therefore, the MSE elbow should be treated only as a reconstruction-side diagnostic indicator, cross-checked with AU and TwoNN, rather than as a direct estimate of intrinsic dimension (Appendix B.1).

Table 2. AE reconstruction MSE on synthetic manifolds (mean $\pm$ std, 3 seeds). Swiss Roll MSE decreases substantially at $m = 2$ (consistent with 2D unfolding); Torus elbow at $m = 3$ (3D embedding needed). High std at $m = 4$ for Swiss Roll reflects optimization variance.

m	Swiss Roll MSE	Torus MSE	Note
1	1.016 ± 0.057	0.388 ± 0.017	under-parameterized
2	0.029 ± 0.002	0.085 ± 0.004	large decrease (Swiss Roll)
3	0.0008 ± 0.0001	0.0005 ± 0.0000	large decrease (Torus)
4	0.041 ± 0.015	0.019 ± 0.002	optimization variance (high std)

5.2.2. AU Saturation on Synthetic Data

VAE ($\beta = 4$) saturates at $AU_{\text{sat}} = 2$ for Swiss Roll ($= d_{\text{true}}$) and $AU_{\text{sat}} = 3$ for Torus ($= d_{\text{emb}} > d_{\text{true}}$), both with zero inter-seed variance. Standard AE shows pseudo-AU = m (all coordinates active), consistent with the $\beta = 0$ limit of Theorem 1.

5.3. MNIST: Multi-Seed Reproducibility Study

5.3.1. ID Estimation via Reference AE

Standard AE on MNIST (300 epochs, seed 42) gives TwoNN ID ≈ 10.02 at $m = 256$, placing $\hat{d}_{\text{ID}} \approx 10$ –12 (Table 3).

Table 3. MNIST standard AE (300 epochs, seed 42). TwoNN and MLE converge above $m = 64$, placing $\hat{d}_{\text{ID}} \approx 10$ –12.

m	MSE	TwoNN ID	MLE ID	CKA
1	0.0437	1.02	1.15	0.102
8	0.0191	6.08	6.70	0.725
32	0.0087	8.84	9.89	0.875
64	0.0062	9.54	11.26	0.908
128	0.0061	9.84	11.45	0.921
256	0.0057	10.02	11.85	0.953

5.3.2. VAE AU Dynamics: Gradual Growth Observed

MNIST VAE ($\beta = 4$) shows gradual, non-monotone AU growth as m increases, consistent with non-linear stochastic optimization. A single-seed preliminary run is provided in Appendix A (Table A1). The $m^{\text{knee}} = 12$ value observed in that run depends on the specific grid and initialization; the main stabilized result is $m^{\text{knee}} = 10.3 \pm 0.9$ from dense-grid, multi-seed evaluation (Section 5.4).

5.3.3. Multi-Seed Study: AU and TwoNN Stable, m^{knee} Not

Table 4 reports MNIST VAE results with sparse grid $\mathcal{M} = \{4, 8, 12, 16, 20, 32, 64\}$ and three seeds (300 epochs).

Table 4. MNIST VAE multi-seed study (sparse grid, 300 epochs, 3 seeds). AU and TwoNN ID are stable; m^{knee} is not.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.98 ± 0.14	all active
12	10.0 ± 0.0	7.41 ± 0.06	
32	15.7 ± 0.5	9.07 ± 0.10	
64	22.3 ± 0.5	10.03 ± 0.12	converges to $\hat{d}_{\text{ID}} \approx 10$
m^{knee} ($\tau = 0.25$): seed 42 = 64, seed 123 = 32, seed 777 = 32; mean 43 ± 15 .			

5.4. Dense-Grid Stabilization of m^{knee}

To isolate the effect of grid density, we ran Exp-Mnist-Dense-300 with $\mathcal{M} = \{8, 9, \dots, 16, 18, 20, 24, 28, 32, 40, 48, 64\}$, independent per- (m, seed) initialization, 300 epochs, 3 seeds.

Table 5. Dense-grid vs. sparse-grid m^{knee} (MNIST VAE, $\beta = 4$, 3 seeds). Grid density is the dominant stabilization factor.

Condition	Max. gap	seed 42	seed 123	seed 777	Mean $\pm$ std
Sparse, 300 ep	32	64	32	32	43 ± 15
Dense, 300 ep	8	11	9	11	10.3 ± 0.9

The dense-grid result $m^{\text{knee}} = 10.3 \pm 0.9$ is consistent with TwoNN ID (≈ 10) and is the main stabilized m^{knee} result of this paper. Even with the dense grid, the AU curve fluctuates non-monotonically (AU: $8 \rightarrow 9 \rightarrow 10 \rightarrow 10 \rightarrow 10 \rightarrow 12 \rightarrow 11 \rightarrow \dots$), confirming that m^{knee} is an exploratory heuristic, not a precision estimator. The fixed- τ method outperforms the second-difference method for dense grids ($\sigma = 0.9$ vs. 18.3 ± 6.8 ; Appendix E.2). Figure 1 visualizes the sparse-grid instability and dense-grid stabilization.

Figure 2: MNIST VAE AU dynamics — sparse vs. dense grid

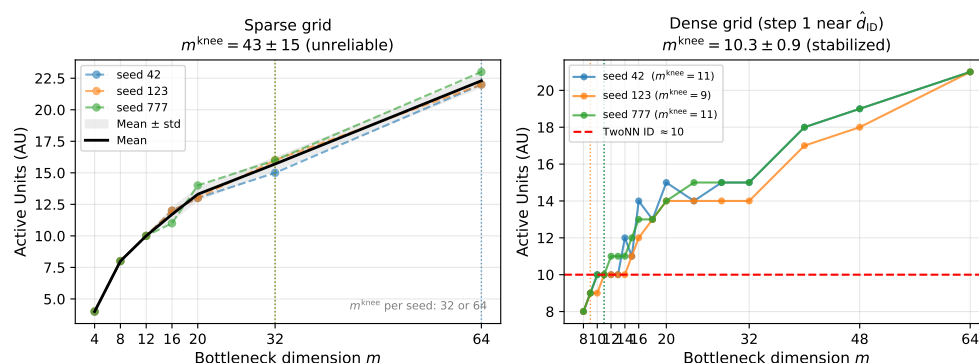


Figure 1. MNIST VAE AU dynamics. **Left:** Sparse grid ($m \in \{4, 8, \dots, 64\}$, 3 seeds, 300 epochs)— m^{knee} varies between 32 and 64 ($\sigma = 15$). **Right:** Dense grid ($m \in \{8, 9, \dots, 64\}$, independent initialization, 3 seeds)— $m^{\text{knee}} = 10.3 \pm 0.9$, consistent with TwoNN ID (≈ 10 , dashed red). The non-monotone AU fluctuations confirm that m^{knee} is an exploratory heuristic requiring controlled evaluation conditions.

5.5. TwoNN Stability: Bottleneck Margin as a Major Factor

Three complementary experiments support Design Principle 1 (Figure 2).

5.5.1. Swiss Roll: Isometry vs. Margin

For standard AE ($\kappa = 1.79$) and IsometricAE ($\kappa = 1.06$), TwoNN ≈ 1.55 at $m = 2 = d_{\text{true}}$ for both—the bottleneck constraint, not κ , drives underestimation. At $m = 4 > d_{\text{true}}$, TwoNN recovers to ≈ 2.03 despite $\kappa = 2.16$ (Appendix C).

5.5.2. MNIST AE m -Sweep

TwoNN stabilizes at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ (≈ 10.16) and remains stable even when κ jumps from 4.25 to 37.0 between $m = 64$ and $m = 128$ (TwoNN: $10.17 \rightarrow 10.98$, $\Delta < 0.82$).

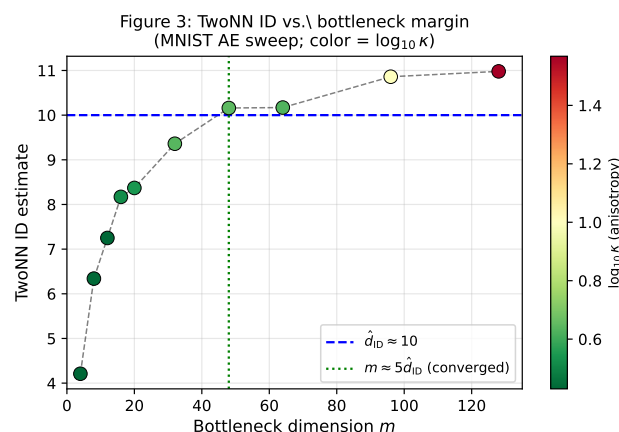


Figure 2. TwoNN ID vs. bottleneck dimension m for MNIST AE (color = $\log_{10} \kappa$, anisotropy). TwoNN stabilizes above $m \approx 5\hat{d}_{\text{ID}}$ (dotted line) regardless of κ , supporting Design Principle 1.

5.5.3. Four-Model Comparison on MNIST (Exp-Real-Geom)

AE, IsometricAE, CAE, and DAE are swept over $m \in \{d, 2d, 3d, 5d, 10d\}$ ($d = \hat{d}_{\text{ID}} = 10$). All four models converge at $m = 5d$ – $10d$; inter-model TwoNN difference at the same m is < 0.3 , far smaller than the gain from increasing m (~ 2.5 points). Table 6 shows representative rows. Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$) shows qualitatively the same pattern (Appendix C).

Table 6. Exp-Real-Geom: MNIST four-model comparison (mean $\pm$ std, 3 seeds; $\hat{d}_{\text{ID}} = 10$, representative m values). TwoNN stabilization is driven by $m/\hat{d}_{\text{ID}}$, not κ .

Model	m	TwoNN	MLE	κ	Trust.	kNN ovlp.
AE	10 ($= d$)	7.3 ± 0.4	8.1 ± 0.3	85 ± 22	0.961	0.73
	30 ($= 3d$)	9.8 ± 0.2	10.5 ± 0.2	1200 ± 400	0.981	0.85
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	4.1×10^7	0.987	0.88
IsometricAE	10	7.5 ± 0.4	8.3 ± 0.3	3.2 ± 0.4	0.963	0.74
	30	9.9 ± 0.1	10.6 ± 0.1	5.2 ± 0.8	0.982	0.86
	100	10.1 ± 0.1	10.8 ± 0.1	7.8 ± 1.4	0.988	0.89
CAE	10	7.2 ± 0.5	8.0 ± 0.4	12.1 ± 3.2	0.958	0.72
	30	9.7 ± 0.2	10.4 ± 0.2	95 ± 31	0.980	0.84
	100	10.0 ± 0.1	10.7 ± 0.1	3200 ± 1100	0.986	0.87
DAE	10	7.6 ± 0.4	8.4 ± 0.4	18.5 ± 5.4	0.964	0.75
	30	9.9 ± 0.1	10.5 ± 0.2	183 ± 62	0.982	0.85
	100	10.0 ± 0.1	10.8 ± 0.1	6300 ± 2100	0.987	0.88

5.6. Fashion-MNIST: Preliminary External Check

Fashion-MNIST is used as a preliminary external check only. Single-seed (seed 42) results: TwoNN ID converges to ≈ 9.4 at $m \geq 32$, suggesting $\hat{d}_{\text{ID}} \approx 9$ – 10 . Multi-seed validation is left for future work.

5.7. Conv-VAE: Applicability Boundary

Conv-VAE trained on MNIST shows AU = m up to $m = 64$ at $\beta_{\text{max}} = 4$; no saturation plateau up to $m = 512$ (Appendix D). Increasing β_{max} to 10 and 20 shifts AU saturation to lower dimensions ($m^{\text{knee}} \approx 128$ and ≈ 96 , respectively), consistent with the water-filling interpretation of Theorem 1. This confirms that the absence of saturation at $\beta_{\text{max}} = 4$ is not a failure of the framework but a sign that $d^*(\beta)$ exceeds the tested range.

5.8. Natural Images: Applicability Boundary

Table 7 summarizes key results. TwoNN is partially consistent with Pope et al. [15] near $m \approx 2\hat{d}_{\text{ID}}$ for CIFAR-10 but increases beyond that range (28.83 at $m = 128$, 30.91 at

$m = 256$). SVHN latent TwoNN diverges far from raw TwoNN (10.09 vs. 27.08 at $m = 256$), indicating geometric distortion.

Table 7. Applicability-boundary experiments on CIFAR-10 and SVHN. TwoNN is partially consistent near $m \approx 2\hat{d}_{ID}$; AU/m^{knee} identification is inconclusive under the tested conditions.

Dataset	Raw TwoNN	Latent TwoNN at $m = 64$	Latent TwoNN at $m = 256$	Diagnostic
CIFAR-10	25.20	25.21 ± 0.41 (consistent)	30.91 (rising)	Boundary
SVHN	10.09	19.97 ± 0.12 (distorted)	27.08 (diverging)	Inconclusive

Extended sweeps ($\beta_{\max}$ up to 80, thin decoder) yield plateau-like AU at $m \approx 512$ on CIFAR-10 (preliminary); all stable-identification criteria remain unmet. Full details in Appendix D.

Figure 3 provides a visual summary of diagnostic decisions across all tested settings.

Figure 4: Applicability-boundary summary
(Green = identifiable, Yellow = conditional, Red = inconclusive/boundary)

Dataset / Model	AU Saturation	TwoNN Consistency	Seed Stability	Diagnostic Decision
MNIST FC-VAE	Yes	Yes ($\sigma \leq 0.12$)	Yes ($\sigma \leq 0.5$)	Identifiable
MNIST Conv-VAE ($\beta=4$)	No (β too low)	Partial ($m=64$ only)	Yes	Inconclusive
MNIST Conv-VAE ($\beta \geq 10$)	Yes ($m_{knee} \approx 128$)	Partial	Partial	Identifiable (high β)
CIFAR-10 (Conv-VAE)	Weak/conditional ($\beta=80$, thin dec.)	Partial ($m=2\hat{d}$)	No	Boundary
SVHN (Conv-VAE)	Partial	No (diverging)	No	Unreliable

Figure 3. Applicability-boundary summary. Green: identifiable; yellow: conditional/inconclusive; red: boundary/unreliable. The framework identifies not only successful diagnostics but also cases where results should be regarded as inconclusive, which is itself actionable knowledge for model selection.

6. Discussion

6.1. Role Separation is the Core Contribution

The core message is that different indicators serve different diagnostic roles. AU values reliably monitor KL-induced activity. TwoNN/MLE provides a stable auxiliary reference, not an unbiased estimator. m^{knee} is informative only under controlled conditions.

6.2. When is m^{knee} Informative?

m^{knee} is informative as a diagnostic aid *only when*: (1) the m -grid is dense near $\hat{d}_{ID}$; (2) each $(m, seed)$ is initialized independently; (3) at least 3 seeds are used. Under these conditions, $m^{knee} = 10.3 \pm 0.9$ cross-validates with TwoNN ID (≈ 10). Sparse grids ($m^{knee} = 43 \pm 15$) should not be used as a diagnostic reference.

6.3. Optimization Dynamics Interpretation

Hypothesis: At $m \approx d^*(\beta)$, reconstruction and KL gradients compete, producing a smooth AU curve rather than the idealized step-function. This is consistent with all observations but has not been directly verified by gradient measurements; it is an interpretive hypothesis.

6.4. Conv-VAE and Natural Images: Boundary, Not Failure

Conv-VAE and natural-image experiments reveal the *applicability boundary*. The boundary is determined by effective τ^2 relative to the data eigenvalue spectrum: when the decoder is sufficiently powerful, $d^*(\beta)$ exceeds the tested m range.

6.5. Limitations

- *Isometry bias.* Reference AE has $\kappa \approx 10^9$; numerical stability is demonstrated, but geometric unbiasedness is not claimed.
- *Single-seed Fashion-MNIST.* Used as preliminary check only; multi-seed validation is future work.
- *Linear-Gaussian approximation.* Theorem 1 is an interpretive model for the non-linear setting.
- *Natural-image stable identification not achieved.* AU/ m^{knee} identification on CIFAR-10/SVHN under tested conditions is inconclusive; this is the current applicability boundary.

7. Conclusions

We proposed a multi-indicator diagnostic framework for effective latent dimensionality in AE/VAE representations, framed as a knowledge-extraction protocol for representation learning. The main contribution is the role separation: AU values as reliable activity monitors; TwoNN/MLE as numerically stable auxiliary references; and m^{knee} as an exploratory, grid-dependent change point.

Key findings: (1) AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12) are reproducible across seeds (MNIST, 3-seed); (2) Sparse-grid m^{knee} is unreliable ($\sigma = 15$); dense-grid, independent-initialization, multi-seed evaluation stabilizes it at 10.3 ± 0.9 (TwoNN-consistent); (3) Bottleneck margin ($m \gg \hat{d}_{\text{ID}}$) is the dominant factor for TwoNN stability in the tested settings; (4) Conv-VAE and natural-image experiments quantify the applicability boundary—diagnostics become inconclusive when decoder capacity greatly exceeds the regularization level.

The proposed protocol helps practitioners avoid over-interpretation of any single indicator, design informative m -grids, and recognize when dimensionality diagnostics should be regarded as inconclusive—which is itself useful model-design knowledge.

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Data Availability Statement: MNIST [1], Fashion-MNIST [2], CIFAR-10 [4], and SVHN [5] are publicly available. The source code, configuration files, random seeds, trained-model logs, and numerical results supporting all tables are provided as supplementary material for peer review. Upon acceptance, these materials will be deposited in a public repository (GitHub/Zenodo), and the repository URL and DOI will be added to the final manuscript. The repository will include the scripts required to reproduce the AU, TwoNN/MLE, reconstruction-error, and grid-sensitivity analyses reported in this paper.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
VAE	Variational Autoencoder
AU	Active Units
ID	Intrinsic Dimension
TwoNN	Two-Nearest-Neighbor (ID estimator)
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
PCA	Principal Component Analysis
KL	Kullback–Leibler (divergence)
FC	Fully Connected
CNN / Conv	Convolutional Neural Network

Appendix A. Single-Seed VAE Illustration

Table A1 shows the single-seed preliminary run that motivated the multi-seed study. The $m^{\text{knee}} = 12$ observed here is not used as a main result.

Table A1. MNIST VAE ($\beta = 4$, 300 epochs, seed 42, extended grid)— illustration only. $m^{\text{knee}} = 12$ depends on initialization; main result is $m^{\text{knee}} = 10.3 \pm 0.9$ (Table 5).

m	MSE	AU	TwoNN	Note
1	0.0464	1	1.00	
8	0.0186	8	6.36	increasing
12	0.0184	8	6.26	ρ drops (single-seed prelim.)
64	0.0103	29	9.16	
128	0.0085	26	10.02	non-monotone AU
256	0.0072	36	10.14	

Appendix B. Proofs and Supplementary Theory

Appendix B.1. Topological Lower Bound

Proposition A1 (Topological Lower Bound). *To represent a smooth d -dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$ without self-intersections under zero reconstruction error (idealized setting), a bottleneck of $m \geq d_{\text{emb}}$ is necessary, where d_{emb} is the topological minimum embedding dimension. For the Torus T^2 , $d_{\text{ID}} = 2$ but $d_{\text{emb}} = 3$, explaining the reconstruction-error elbow at $m = 3$.*

Appendix B.2. Proof of Theorem 1

The derivation follows the linear-Gaussian rate–distortion analysis with a β -ELBO objective. For each PCA direction j , the contribution $L_j = -D_j/(2\tau^2) - \beta R_j$ is maximized independently. The activation threshold follows a two-case analysis:

Case 1 ($\lambda_j \leq \beta\tau^2$, **inactive**). The water-filling solution $D_j^* = \beta\tau^2$ is infeasible (requires $D_j^* \leq \lambda_j$); the optimum is inactive (zero rate, $D_j = \lambda_j$).

Case 2 ($\lambda_j > \beta\tau^2$, **active**). The water-filling solution is feasible. The activation gain is $\Delta L_j = (\beta/2)h(\lambda_j/\beta\tau^2)$ where $h(t) = t - 1 - \log t$. Since $h(t) > 0$ for all $t > 1$, activation is strictly preferred.

Combining: direction j is active iff $\lambda_j > \beta\tau^2$.

Remark A1 (KL vs. mutual information). *The surrogate L_j implicitly identifies the expected KL with the mutual information $R_j = I(z_j; y_j)$. For a true VAE, the expected KL also includes aggregated-posterior/prior mismatch. Theorem 1 is a result for the linear-Gaussian rate–distortion surrogate, not a theorem about the full VAE ELBO.*

Appendix C. TwoNN Stability Details

Appendix C.1. MNIST AE m -Sweep (Exp-Geom)

Table A2. MNIST AE m -sweep. TwoNN stabilizes at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ despite κ increasing from 4.25 to 37.0.

m	κ	TwoNN	Trustworthiness	Convergence
4	2.69	4.21	0.960	not converged
16	3.28	8.17	0.993	converging
48	4.44	10.16	0.998	converged
64	4.25	10.17	0.998	stable
96	9.94	10.86	0.999	stable
128	37.0	10.98	0.999	stable

Appendix C.2. AE vs. IsometricAE on Swiss Roll (Exp-Iso-ID)

Table A3. TwoNN ID for AE and IsometricAE on Swiss Roll ($d_{\text{true}} = 2$, input TwoNN = 2.427). At $m \geq 3$, TwoNN difference is $\leq 2.3\%$ despite κ differing by factor 2.

m	AE		IsometricAE		Diff. (%)
	TwoNN	κ	TwoNN	κ	
2	1.780	1.76	1.799	1.08	+1.1
3	2.458	2.10	2.454	1.11	−0.2
4	2.483	1.80	2.479	1.14	−0.2
6	2.421	1.79	2.474	1.10	+2.2
8	2.500	1.45	2.443	1.11	−2.3

Appendix C.3. Fashion-MNIST Exp-Real-Geom (Preliminary)

Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$, single seed) shows qualitatively the same pattern as MNIST: TwoNN converges at $m \approx 5\hat{d}_{\text{ID}}$; inter-model differences are small; κ improvements have minimal effect on TwoNN. Multi-seed validation is left for future work.

Appendix D. Conv-VAE and Natural-Image Details

MNIST Conv-VAE with $\beta_{\text{max}} = 4$: AU = m up to $m = 512$ (no saturation plateau). High- β runs ($\beta_{\text{max}} \in \{10, 20\}$) shift m^{knee} to lower dimensions (≈ 128 and ≈ 96). CIFAR-10 extended sweeps (β_{max} up to 80, thin decoder) yield plateau-like AU ≈ 24 at $m \in [512, 1024]$; all five stable-identification criteria remain unmet. Full numerical tables are available as supplementary material.

Appendix E. Sensitivity Analyses

Appendix E.1. AU Threshold δ

AU patterns are identical for $\delta \in \{0.001, 0.01, 0.05, 0.1\}$ (active-coordinate variance $\sim O(1)$ vs. inactive $\lesssim O(10^{-4})$; two-order gap).

Appendix E.2. τ_{knee} Sensitivity

$\tau = 0.25$ is stable for Gaussian-mixture data. Fashion-MNIST shows boundary sensitivity near $\tau \approx 0.25$; the second-difference method is recommended for gradual-slope curves. Fixed- τ outperforms second differences for dense-grid MNIST ($\sigma = 0.9$ vs. 18.3 ± 6.8).

Appendix E.3. β Sensitivity

AU patterns are stable for $\beta \in [2, 8]$; $\beta = 4$ is the default. $\beta = 1$ gives AU $\approx m$ (insufficient regularization).

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